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Project Title:	Counteracting Space Radiation by Targeting Neurogenesis in a Human Brain Organoid Model		
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Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
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Comments:			
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Start Date:	10/01/2020	**End Date:**	09/30/2023
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Grosshans, David M.D., Ph.D. (The University of Texas M.D. Anderson Cancer Center) Gaber, Mostafa Ph.D. (Baylor College of Medicine) Young, Damian Ph.D. (Baylor College of Medicine) Liu, Zhandong Ph.D. (Baylor College of Medicine)		
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Task Description:	Neurogenesis, the generation of new neurons throughout life, is essential for formation of new spatial memory and mood control in the hippocampus. New neurons are formed from neural stem cells, which are very sensitive to all forms of radiation. If exposed, they die and therefore, neurogenesis declines leading to decline in learning and memory as well as depression. Therefore, understanding this phenomenon in the context of space radiation is of utmost importance if we are to avoid at least some of the cognitive and mental health pathologies during spaceflight. Herein, we propose to examine neurogenesis in the human brain organoid models exposed to Linear Energy Transfer (LET) proton beam to mimic Galactic Cosmic Rays (GCR). These models are ideal not only to examine the effects of radiation on neural stem cells but also the effects of drug compounds, because they provide a high-throughput system with multiple neurogenic sites (rosettes) and diverse cell types including both neurons and astrocytes. We will use two complementary cerebral organoid models and will expose them to the proton beam at the MD Anderson Proton Center at different time points and different frequency of exposure and will examine molecular, metabolic, cellular, and physiological properties of the variety of cell types that are part of the neurogenic niche. Furthermore, we will take advantage of our new small molecules that promote neural stem cell self-renewal and neurogenesis in animal models in vivo. Based on their mechanism of action, they may also decrease microglial inflammation; thus, targeting multiple elements of the neurogenic niche. In addition, we will use transient bursts of electrical stimulation as a non-pharmacological countermeasure to GCR radiation, as increased neuronal activity promotes neurogenesis. Overall, our proposal is responsive to all requirements of this solicitation and stands to deliver new data relevant to the effects of space radiation on multiple human cell types that are part of the neurogenic niche and the efficacy of countermeasures on molecular, metabolic, cellular, and physiological properties of these cells. Our work focuses on neurogenesis as this is the only natural mechanism to regenerate lost brain tissue in vivo in the center for learning and memory and mood control. Thus, the relevance to this solicitation and NASA objective in general is high.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal arises from our pioneering studies on adult neurogenesis. We have long known that low levels of neurogenesis are associated with cognitive decline and depression in humans and animal models. However, because no one could measure this process in the human brain in vivo, discovery in this critical area has been stagnant. Our team was the first to demonstrate that neural stem cells (NSCs) that generate newborn neurons are abundant in mono-unsaturated fatty acids (MUFAs). These fatty acids are detectable both in vitro and in vivo, including in the human brain. We also showed that 18:1?9 MUFAs (oleic acid) bind to tailless receptor (TLX) (a nuclear receptor critical for NSC proliferation and neurogenesis) and that this binding increases NSC proliferation and neurogenesis. We now use this knowledge to counteract the detrimental space radiation effects on NSCs and neurogenesis. To accomplish this goal and, eventually, develop efficient clinical interventions, we use the most innovative chemistry, biophysics, molecular imaging, and complex human brain organoid models available to determine the cellular, molecular, metabolic, and neuroimmune effects of space-like radiation on NSCs and neurogenesis and then counteract with potent stimulators of neurogenesis. To stimulate NSCs and neurogenesis, we engaged novel synthetic chemistry to identify small molecules that can modulate the function of TLX and stimulate the generation of newborn neurons, creating a paradigm shift in our quest to achieve cell-based repair and brain regeneration in the future. To this end, we will employ a highly innovative platform developed in Dr. Young's lab to generate potent and selective drugs based on Fragment-Based Drug Discovery (FBDD). As an alternative strategy to high-throughput screening (HTS) of large screening collections, FBDD focuses on the development of optimal ligands. FBDD originated in the early 1990s when Jencks found that low molecular weight (MW) and weak-binding ligands to protein targets are actually highly efficient binders. A decade later, medicinal chemists began to appreciate that these "fragments" were valuable starting points for developing lead-like compounds. Fragments are generally defined as small molecules with a MW<300Da that bear fewer than 20 heavy atoms. Although fragments bind to their targets with low affinity, their ligand efficiency (defined as the free energy of binding per heavy atom) is high. Given this thermodynamic efficiency metric, fragment hits—which are ideally contoured to the small area in which they bind to a given protein—are nearly optimal binders. We can convert the weak-binding fragment into a more potent modulator by using synthetic organic chemistry to opportunistically place substituents or "grow" the fragment to increase productive interactions within the binding pocket. Fragments' smaller size also allows us to explore the chemical space they occupy more efficiently. Accordingly, a library of several thousand appropriately chosen fragments is usually sufficient to yield high-quality binders for a particular target rather than the millions of compounds required for HTS. Fragment-screening is generally achieved using highly sensitive biophysical experiments capable of detecting weak fragment-target binders. With FBDD, however, we need fewer compounds and can use biophysical approaches to develop versatile screening platforms, enabling us to produce results in a more efficient, cost-effective way. The Young lab has synthetically constructed a highly 3D-diverse fragment library toward the goal of modulating a wide swath of biological targets. These fragments have already shown to be promising starting points for developing novel TLX ligands, and thus, stimulating NSCs to produce new neurons when needed.
	Neurogenesis, the generation of new neurons throughout life, is essential for formation of new spatial memory and mood control. New neurons are formed from neural stem cells, which are very sensitive to all forms of radiation. If exposed, they die; and therefore, neurogenesis declines, leading to decline in learning and memory as well as depression. Thus, understanding this phenomenon in the context of space radiation is of utmost importance if we are to avoid at least some of the cognitive and mental health pathologies during long-term space flights. In our proposal, we examined neurogenesis in the human brain organoid models exposed to a proton beam to mimic Galactic Cosmic Ray (GCR) irradiation. These models are ideal not only to examine the effects of radiation on neural stem cells but also the effects of countermeasures, because they provide a high-throughput system with multiple neurogenic sites (rosettes) and diverse cell types including both neurons and astrocytes. We exposed organoid models to the proton beam at the MD Anderson Proton Center at different time points and different frequency of exposure. In Aim 1, we examined macroscopic effects of GCR-like radiation on brain organoids. In Aim 2, we established the microscopic effects of GCR-like radiation on brain organoids; and in Aim 3, we tested the effects of our new small molecules that target TLX/NR2E1, a nuclear receptor that promotes neural stem cell self-renewal and neurogenesis in animal models in vivo. Over the course of the grant, we have made major accomplishments in our proposed experiments. We discovered that a single exposure of 30-day-old organoids (when they mostly contain neuroprogenitors) to 0.5 Gy proton beam leads to increased apoptosis and decreased cell cycle within the first 24hr of irradiation. At that time, irradiation mostly leads to

Task Progress:	neural stem cell apoptosis; while by day 3, neuronal death is more abundant. By 10 days following irradiation, the organoid tissue normalizes and there is no significant difference in apoptosis, cell cycle, transcriptome (bulk RNAseq), and metabolome/lipidome between irradiated and non-irradiated organoids. Organoids, however, progressively decrease in size, and this becomes significant at 10 days post-irradiation and persists up to 30 days (the latest time point we analyzed). scRNAseq confirmed our results from the bulk RNAseq and further established that, at day 1, most affected cells were neural stem cells; while at day 3, most affected cells were neurons. Gene Ontology analyses confirmed that metabolism, specifically glycolysis and lipid metabolism, was mostly affected by irradiation. RNA velocity analysis, which calculates the ratio between unspliced and spliced isoforms of RNAs to indicate whether a cell is changing from one type to another (e.g., as result of cell differentiation), suggested that irradiation prolongs the differentiation time of neuroblasts. Further, using a fragment-based drug discovery approach, we developed a set of synthetic molecules that bind to the nuclear receptor TLX/NR2E1 (the master regulator of neurogenesis) as agonists. Out of about 30 generated compounds, we determined two that had the most robust effect on neurogenesis in healthy organoids. 2uM of these drugs, given every two days for a week, was not toxic and led to increased gene expression of NSC and newborn neuron genes. When we treated the organoids before irradiation, the relative size of irradiated organoids at 10 days was larger compared to non-treated irradiated organoids, suggesting that our synthetic compounds may stimulate remaining NSCs in the organoids to produce more neurons and recover from the injury. Finally, we also developed a new algorithm for identification of metabolic networks critical for neurogenesis. This algorithm combines multivariate principal component analyses (PCA) and graphical lasso (GLASSO) to capture both metabolome differences over time and the functional pathway modules that contribute to these differences. Overall, our work delivered new data relevant to the effects of space-like radiation on multiple human cell types that are part of the neurogenic niche and the efficacy of countermeasures on molecular, metabolic, cellular, and physiological properties of these cells.
Bibliography Type:	Description: (Last Updated: 11/06/2025)
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